PROJECT REVIEW

SOCIETY MANAGEMENT PORTAL

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Under the Guidance of
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GROUP NUMBER: 18

Name of Candidate	Role
Sagi Sahasra Srihasa	Frontend Developer
Pamarthy Tejaswini	Database and Backend Developer

- Both members contributed to integration, testing and project documentation.
- The completed system uses a React frontend connected to an Express/Node.js backend and MySQL database.

ABSTRACT

• The Society Management Portal is a full-stack web application for managing residential society activities through a centralized platform.

• It provides separate Admin and Resident interfaces for managing residents, payments, visitors, complaints, amenities, notices, polls and notifications.

• The application uses REST APIs to connect the React frontend with the Node.js/Express backend and MySQL relational database.

• The completed implementation replaces the earlier frontend-only/localStorage approach with persistent backend database storage.

INTRODUCTION

• Residential societies handle many activities such as resident records, maintenance payments, visitors, complaints, amenities and announcements.

• Managing these activities separately can make records difficult to maintain and information difficult to access.

• Our portal brings these functions into one web-based application with role-based Admin and Resident interfaces.

• The system provides a common database so information can be stored and retrieved consistently.

EXISTING SYSTEM

- Society activities may be maintained using registers, spreadsheets or separate communication channels.
- Records can become scattered across different sources.
- Tracking payments, complaints, visitors and notices manually can take additional time.
- Residents and administrators may not have a single place to view the latest information.
- A centralized database-backed application can organize these activities in one system.

PROBLEM

- Lack of a centralized platform for society operations.
- Difficulty maintaining and accessing resident and flat records.
- Limited tracking of payment status and other society activities.
- Manual handling of complaints, visitors and amenity-related requests.
- Communication gaps when notices and notifications are maintained separately.

SOLUTION / CONTRIBUTION

- Developed a centralized Society Management Portal with separate Admin and Resident interfaces.
- Implemented a React frontend connected to a Node.js and Express REST API backend.
- Used MySQL to persist users, residents, payments, visitors, complaints, amenities, notices, polls and notifications.
- Implemented resident registration/login and password hashing using bcryptjs.
- Implemented resident deletion from the Admin side so moved-out residents can be removed from the system.
- Connected the main Admin and Resident modules to backend APIs and database data.

OBJECTIVES

- Develop a centralized platform for residential society management.
- Digitize common society operations using a web application.
- Maintain organized resident and flat information in a relational database.
- Provide payment, visitor and complaint management features.
- Manage amenities, notices and society polls systematically.
- Provide notifications for important updates.
- Connect the frontend, backend APIs and MySQL database into one working system.

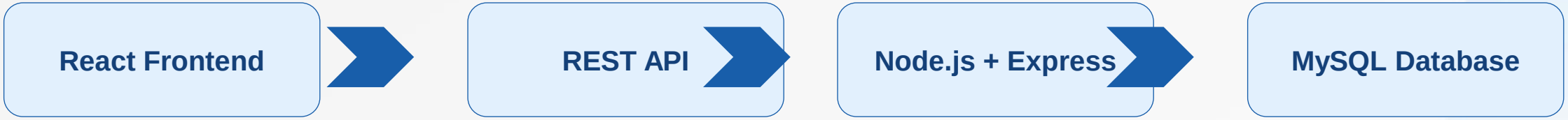
MODULES

Module	Implemented functionality
Admin	Dashboard, residents, payments, visitors, complaints, amenities, notices, polls, notifications
Resident	Dashboard, profile, payments, visitors, complaints, amenities, notices, polls, notifications
Authentication	Admin/Resident login, resident signup, password hashing
Database	MySQL tables for users, roles, residents and society activities

SOFTWARE REQUIREMENTS / TECHNOLOGIES USED

Layer	Technology actually used
Frontend	React.js, JavaScript, HTML, CSS
Backend	Node.js, Express.js
Database	MySQL
API communication	REST API / Fetch
Authentication	bcryptjs password hashing
Development	Visual Studio Code
Version control	Git & GitHub

SYSTEM ARCHITECTURE, DATABASE & OUTCOME



- Core database tables include roles, users, residents, payments, notifications, notices, complaints, amenities, polls and visitors.
- The completed portal successfully demonstrates persistent database-backed management instead of relying only on frontend dummy data.
- The system provides a working base for future mobile support, automated reminders, reports and additional society services.

CONCLUSION AND FUTURE SCOPE

- The Society Management Portal centralizes important residential society activities in a single full-stack application.
- The project integrates React, Node.js/Express REST APIs and MySQL to provide persistent and organized data management.
- Separate Admin and Resident interfaces make the system suitable for different user responsibilities.
- Future scope includes a dedicated mobile application, automated reminders, richer reports/analytics, payment-gateway integration and additional access-control features.

THANK YOU